Weighted Sum-Rate Maximization using Weighted MMSE for MIMO-BC Beamforming Design

Myoungjin Oh

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School of Electrical Engineering, Korea University

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I. Introduction

- For MIMO-BC, non-linear techniques outperform linear ones.
  - However, they are complex for both Tx and Rx.
- Focus on Tx Beamforming design to **Maximize Weighted Sum-Rate (WSR)**.
  - WSR maximization is non-convex and non-trivial.
- Establish a relationship between WSR maximization and **Weighted Minimum Mean Square Error (WMMSE)** minimization.
- Propose an iterative algorithm to solve WSR maximization via WMMSE minimization.

System Model

General narrow band point-to-multipoint MIMO system.

- P transmit antennas
- K users each with Q receive antennas
- System has total QK receive antennas
- MIMO Channel between Tx and user k : $\mathbf{H}_k \in \mathbb{C}^{[Q\times P]}$

$$\mathbf{y}_k(n) = \mathbf{H}_k\mathbf{x}(n) + \mathbf{v}_k(n), \tag{1}$$

Where

- $\mathbf{x}(n) \in \mathbb{C}^{[1\times P]}$: complex transmitted vector
- $\mathbf{v}_k(n) \in \mathbb{C}^{[Q\times 1]}$: noise vector containing circularly symmetric white Gaussian noise

System Model

$\mathbf{x}(n)$: linearly filtered version of the input data vectors $\mathbf{d}_1(n), \cdots, \mathbf{d}_K(n) \in \mathbb{C}^{[Q \times 1]}$

$$\mathbf{x}(n) = \sum_{k=1}^{K} \mathbf{B}_k \mathbf{d}_k(n). \tag{2}$$

Matrices $\mathbf{B}_1, \cdots, \mathbf{B}_K \in \mathbb{C}^{[P \times Q]}$: transmit filters(Beamformers). Assumed

- each user has Q parallel data streams.
- each user receives Q independent streams such that $\mathbb{E}[\mathbf{d}_k(n)\mathbf{d}_k(n)^{\mathrm{H}}] = \mathbf{I}_Q$

System Model

Transmit power constraint

$$\frac{1}{N}\sum_{n=1}^{N}\mathbf{x}^{\mathrm{H}}(n)\mathbf{x}(n) \leq E_{tx} \tag{3}$$

If N is large enough, then

$$\frac{1}{N}\sum_{n=1}^{N}\mathbf{x}^{\mathrm{H}}(n)\mathbf{x}(n) \rightarrow \mathbb{E}[\mathbf{x}^{\mathrm{H}}(n)\mathbf{x}(n)] = \sum_{k}\mathrm{Tr}(\mathbf{B}_k\mathbf{B}_k^{\mathrm{H}}) \tag{3a}$$

Derivation

A. Main Objective

To find $\mathbf{B}_1, \cdots, \mathbf{B}_K$ that maximizes WSR under power constraint (3):

$$[\mathbf{B}_1^{\mathrm{WSR}}, \cdots, \mathbf{B}_K^{\mathrm{WSR}}] = \underset{\mathbf{B}_1, \cdots, \mathbf{B}_K}{\arg\min} \sum_k -u_{R_k} R_k \qquad (4)$$

$$\text{s.t.} \sum_{k=1}^{K} \mathrm{Tr}(\mathbf{B}_k \mathbf{B}_k^{\mathrm{H}}) = E_{tx},$$

Where

- $u_{R_k} \geq 0$: weight for user k
- R_k : achievable rate for user k

$$R_k = \log\det\left(\mathbf{I}_k + \mathbf{B}_k^{\mathrm{H}} \mathbf{H}_k^{\mathrm{H}} \mathbf{R}_{\tilde{v}_k \tilde{v}_k}^{-1} \mathbf{H}_k \mathbf{B}_k\right), \qquad (5)$$

$$\mathbf{R}_{\bar{v}_k \bar{v}_k} = \mathbf{I}_k + \sum_{i=1, i \neq k}^{K} \mathbf{H}_k \mathbf{B}_i \mathbf{B}_i^{\mathrm{H}} \mathbf{H}_k^{\mathrm{H}} \qquad (6)$$

A. Main Objective

The MMSE receive filter at user k is given by

$$\begin{aligned}\mathbf{A}_k^{\mathrm{MMSE}} &= \underset{\mathbf{A}_k}{\arg\min}\ \mathbb{E}\left[\left\|\mathbf{A}_k\mathbf{y}_k - \mathbf{d}_k\right\|^2\right] \\ &= \mathbf{B}_k^{\mathrm{H}}\mathbf{H}_k^{\mathrm{H}}\left(\mathbf{H}_k\mathbf{B}_k\mathbf{B}_k^{\mathrm{H}}\mathbf{H}_k^{\mathrm{H}} + \mathbf{R}_{\bar{v}_k\bar{v}_k}\right)^{-1} \end{aligned} \tag{7}$$

Derivation

MSE Matrix for user k given at the MMSE-receive filter

$$\begin{aligned}\mathbf{E}_k &= \mathbb{E}\left[\left(\mathbf{A}_k^{\mathrm{MMSE}}\mathbf{y}_k - \mathbf{d}_k\right)\left(\mathbf{A}_k^{\mathrm{MMSE}}\mathbf{y}_k - \mathbf{d}_k\right)^{\mathrm{H}}\right] \\ &= \left(\mathbf{I}_k + \mathbf{B}_k^{\mathrm{H}}\mathbf{H}_k^{\mathrm{H}}\mathbf{R}_{\tilde{v}_k\tilde{v}_k}^{-1}\mathbf{H}_k\mathbf{B}_k\right)^{-1} \end{aligned} \tag{8}$$

Derivation

A. Main Objective

Given (5) and (8), the rate for user k can be expressed as

$$R_k = -\log\det(\mathbf{E}_k^{-1}) \tag{9}$$

A. Gradient of WSR Maximization

Formulate Lagrangian for WSR maximization problem (4):

$$\{(\mathbf{B}_1, \ldots, \mathbf{B}_K, \lambda) = \underbrace{\sum_{k=1}^{K} -u_{R_k} R_k}_{\text{Objective: Minimize Negative WSR}} + \lambda \underbrace{\left(\sum_{k=1}^{K} \mathrm{Tr}(\mathbf{B}_k \mathbf{B}_k^{\mathrm{H}}) - E_{tx} \right)}_{\text{Constraint: Total Power Limit}} \tag{12}$$

Define $\nabla_{\mathbf{B}_k} f = \frac{\partial f}{\partial \mathbf{B}_k^*}$

A. Gradient of WSR Maximization

$$\nabla_{\mathbf{B}_k} f = \nabla_{\mathbf{B}_k} \left(\sum_{j=1}^{K} -u_j R_j + \lambda \left(\sum_{j=1}^{K} \mathrm{Tr}(\mathbf{B}_j \mathbf{B}_j^{\mathrm{H}}) - E_{tx} \right) \right)$$

$$= -u_k \underbrace{\nabla_{\mathbf{B}_k} R_k}_{\text{(i) Self-Rate}} - \sum_{i \neq k} u_i \underbrace{\nabla_{\mathbf{B}_k} R_i}_{\text{(ii) Interference}} + \lambda \mathbf{B}_k = \mathbf{0}$$

Compute $\nabla_{\mathbf{B}_k} R_k$

$$\text{Definition:} \quad R_k = \log\det\left(\mathbf{I} + \mathbf{B}_k^{\mathrm{H}} \mathbf{H}_k^{\mathrm{H}} \mathbf{R}_{\tilde{v}_k \tilde{v}_k}^{-1} \mathbf{H}_k \mathbf{B}_k\right)$$

$$\text{Gradient:} \quad \nabla_{\mathbf{B}_k} R_k = \mathbf{H}_k^{\mathrm{H}} \mathbf{R}_{\tilde{v}_k}^{-1} \mathbf{H}_k \mathbf{B}_k \underbrace{\left(\mathbf{I} + \mathbf{B}_k^{\mathrm{H}} \mathbf{H}_k^{\mathrm{H}} \mathbf{R}_{\tilde{v}_k \tilde{v}_k}^{-1} \mathbf{H}_k \mathbf{B}_k\right)^{-1}}_{\mathbf{E}_k \text{ (MMSE Matrix)}}$$

$$\therefore \quad \nabla_{\mathbf{B}_k} R_k = \mathbf{H}_k^{\mathrm{H}} \mathbf{R}_{\tilde{v}_k \tilde{v}_k}^{-1} \mathbf{H}_k \mathbf{B}_k \mathbf{E}_k \tag{14}$$

▸ Derivation

A. Gradient of WSR Maximization

Compute $\nabla_{\mathbf{B}_k} R_i$ for $i \neq k$

$$\nabla_{\mathbf{B}_k} R_i = -\mathbf{H}_i^{\mathrm{H}} \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \mathbf{H}_i \mathbf{B}_i \mathbf{E}_i \mathbf{B}_i^{\mathrm{H}} \mathbf{H}_i^{\mathrm{H}} \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \mathbf{H}_i \mathbf{B}_k \tag{17}$$

Derivation

Combine (14) and (17), we obtain the gradient of WSR maximization problem (10):

$$\begin{aligned} \nabla_{\mathbf{B}_k} f &= -u_k \underbrace{\nabla_{\mathbf{B}_k} R_k}_{\text{(i) Self-Rate}} - \sum_{i \neq k} u_i \underbrace{\nabla_{\mathbf{B}_k} R_i}_{\text{(ii) Interference}} + \lambda \mathbf{B}_k \\ &= -u_k \mathbf{H}_k^{\mathrm{H}} \mathbf{R}_{\tilde{v}_k \tilde{v}_k}^{-1} \mathbf{H}_k \mathbf{B}_k \mathbf{E}_k + \left(\sum_{i=1, i \neq k}^{K} u_i \mathbf{H}_i^{\mathrm{H}} \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \mathbf{H}_i \mathbf{B}_i \mathbf{E}_i \mathbf{B}_i^{\mathrm{H}} \mathbf{H}_i^{\mathrm{H}} \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \mathbf{H}_i \right) \mathbf{B}_k + \lambda \mathbf{B}_k \end{aligned} \tag{10}$$

B. Gradient of WMMSE Minimization

The WMMSE minimization problem is given by

$$\left[\mathbf{B}_1^{\mathrm{MMSE}}, \cdots, \mathbf{B}_K^{\mathrm{MMSE}}\right] = \underset{\mathbf{B}_1, \cdots, \mathbf{B}_K}{\arg\min} \sum_k \mathrm{Tr}\left(\mathbf{W}_k \mathbf{E}_k\right) \quad \text{s.t.} \sum_{k=1}^{K} \mathrm{Tr}\left(\mathbf{B}_k \mathbf{B}_k^{\mathrm{H}}\right) = E_{tx} \tag{18}$$

$\mathbf{W}_k \in \mathbb{C}^{[Q_k \times Q_k]}$: constant weight matrix associated with user k.

Lagrangian leads:

$$g\left(\mathbf{B}_1 \cdots \mathbf{B}_K\right) = \sum_k \mathrm{Tr}\left(\mathbf{W}_k \mathbf{E}_k\right) + \lambda \left(\sum_k \mathrm{Tr}\left(\mathbf{B}_k \mathbf{B}_k^{\mathrm{H}}\right) - E_{tx}\right) \tag{19}$$

Compute $\nabla_{\mathbf{B}_k} g$

1 Compute $\nabla_{\mathbf{B}_k} \mathrm{Tr}\left(\mathbf{W}_k \mathbf{E}_k\right)$

$$\nabla_{\mathbf{B}_k} \mathrm{Tr}\left(\mathbf{W}_k \mathbf{E}_k\right) = -\mathbf{H}_k^{\mathrm{H}} \mathbf{R}_{\tilde{v}_k \tilde{v}_k}^{-1} \mathbf{H}_k \mathbf{B}_k \mathbf{E}_k \mathbf{W}_k \mathbf{E}_k \tag{20}$$

Derivation

2 Compute $\nabla_{\mathbf{B}_k} \mathrm{Tr}\left(\mathbf{W}_i \mathbf{E}_i\right)$ for $i \neq k$

$$\nabla_{\mathbf{B}_k} \mathrm{Tr}\left(\mathbf{W}_i \mathbf{E}_i\right) = \mathbf{H}_i^{\mathrm{H}} \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \mathbf{H}_i \mathbf{B}_i \mathbf{E}_i \mathbf{W}_i \mathbf{E}_i \mathbf{B}_i^{\mathrm{H}} \mathbf{H}_i^{\mathrm{H}} \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \mathbf{H}_i \mathbf{B}_k \tag{21}$$

Derivation

3 Combining (20) and (21), we obtain the WMMSE gradient:

$$\begin{aligned}\nabla_{\mathbf{B}_k} g &= \nabla_{\mathbf{B}_k} \mathrm{Tr}\left(\mathbf{W}_k \mathbf{E}_k\right) + \left(\sum_{i=1, i\neq k}^{K} \nabla_{\mathbf{B}_k} \mathrm{Tr}\left(\mathbf{W}_i \mathbf{E}_i\right)\right) + \lambda \mathbf{B}_k \\ &= -\mathbf{H}_k^{\mathrm{H}} \mathbf{R}_{\tilde{v}_k \tilde{v}_k}^{-1} \mathbf{H}_k \mathbf{B}_k \mathbf{E}_k \mathbf{W}_k \mathbf{E}_k \\ &\quad + \left(\sum_{i=1, i\neq k}^{K} \mathbf{H}_i^{\mathrm{H}} \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \mathbf{H}_i \mathbf{B}_i \mathbf{E}_i \mathbf{W}_i \mathbf{E}_i \mathbf{B}_i^{\mathrm{H}} \mathbf{H}_i^{\mathrm{H}} \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \mathbf{H}_i\right) \mathbf{B}_k + \lambda \mathbf{B}_k \end{aligned} \qquad (11)$$

Notice also

$$\nabla_\lambda f = \nabla_\lambda g = \sum_k \mathrm{Tr}\left(\mathbf{B}_k \mathbf{B}_k^{\mathrm{H}}\right) - E_{tx}$$

C. Compare $\nabla_{\mathbf{B}_k}$WSR and $\nabla_{\mathbf{B}_k}$WMMSE

Key Insight: The two problems are closely related through the MSE weights.

- If we select the MSE weights as:

$$\mathbf{W}_k = u_k \mathbf{E}_k^{-1}$$

- Then, the gradients become identical:

$$\nabla_{\mathbf{B}_k}\text{WSR} = \nabla_{\mathbf{B}_k}\text{WMMSE}$$

Implication

A stationary point of the WSR problem satisfies the **KKT conditions** of the WMMSE problem.
$\Rightarrow$ We can solve WSR by iteratively solving WMMSE.

Proposed Algorithm: WSRBF-WMMSE

Algorithm 1: WSRBF-WMMSE Algorithm

```
Set $n = 0$
Set $\mathbf{B}_k^n = \mathbf{B}_k^{\text{init}} \forall k$
while not converged do
    update $n \leftarrow n + 1$
    I. compute $\mathbf{A}_k^n \mid \mathbf{B}_i^{n-1} \forall i$ for all $k$ using (7)
    II. compute $\mathbf{W}_k^n \mid \mathbf{B}_i^{n-1} \forall i$ for all $k$ using (22),(8)
    III. compute $\mathbf{B}^n \mid \mathbf{A}^n, \mathbf{W}^n$ using (23),(24)
end
```

The WMMSE transmit filter structure is computed as:

$$\bar{\mathbf{B}} = \left(\mathbf{H}^{\mathrm{H}}\mathbf{A}^{\mathrm{H}}\mathbf{W}\mathbf{A}\mathbf{H} + \frac{\mathrm{Tr}\left(\mathbf{W}\mathbf{A}\mathbf{A}^{\mathrm{H}}\right)}{E_{tx}}\mathbf{I}_P\right)^{-1}\mathbf{H}^{\mathrm{H}}\mathbf{A}^{\mathrm{H}}\mathbf{W}, \tag{23}$$

Derivation

- $\mathbf{W}_{[QK\times QK]} = \mathrm{diag}\left\{\mathbf{W}_1, \cdots, \mathbf{W}_K\right\}$
- $\mathbf{A}_{[QK\times QK]} = \mathrm{diag}\left\{\mathbf{A}_1, \cdots, \mathbf{A}_K\right\}$
- $\mathbf{H}_{[QK\times P]} = \left[\mathbf{H}_1^{\mathrm{T}}, \cdots, \mathbf{H}_K^{\mathrm{T}}\right]^{\mathrm{T}}$: contains channel matrices stacked row-wise

Proposed Algorithm: WSRBF-WMMSE

$$\bar{\mathbf{B}} = \left(\underbrace{\mathbf{H}^{\mathrm{H}}\mathbf{A}^{\mathrm{H}}\mathbf{W}\mathbf{A}\mathbf{H}}_{\text{Signal Direction}} + \underbrace{\frac{\mathrm{Tr}(\mathbf{W}\mathbf{A}\mathbf{A}^{\mathrm{H}})}{E_{tx}}\mathbf{I}_P}_{\text{Regularization Term}} \right)^{-1} \mathbf{H}^{\mathrm{H}}\mathbf{A}^{\mathrm{H}}\mathbf{W}$$

Regularization Term ($\mu \mathrm{I}_P$):

- The scalar factor $\mu = \frac{\mathrm{Tr}(\mathbf{W}\mathbf{A}\mathbf{A}^{\mathrm{H}})}{E_{tx}}$ acts as a noise-to-signal ratio.
- Unlike standard ZFBF, this term prevents noise amplification when the channel is ill-conditioned.

Behavior at Extremes:

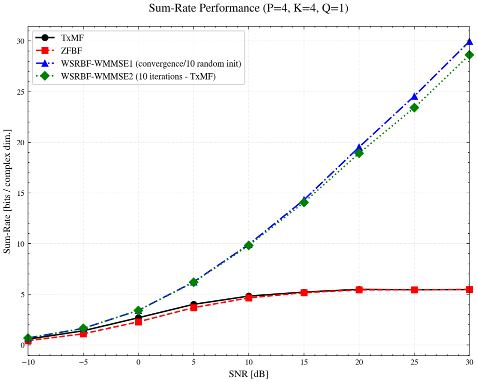
- **High SNR (**$E_{tx} \to \infty$**):** $\mu \to 0$, behaves like **Zero-Forcing (ZFBF)**.
- **Low SNR (**$E_{tx} \to 0$**):** μ dominates, behaves like **Matched Filter (TxMF)**.

V. Simulation

Reproduce Setup

- $\mathbf{H}_k$: Rayleigh fading channel with path loss
- For all users, $u_{R_k} = 1$
- For matched filter (TxMF), $\mathbf{B}_k = \beta \mathbf{H}_k^{\mathrm{H}}$
- For zero-forcing beamforming (ZFBF), $\mathbf{B} = \beta \mathbf{H}^{\mathrm{H}} (\mathbf{H}\mathbf{H}^{\mathrm{H}})^{-1}$
- w/ Python, NumPy

Reprocude Simulation Result



Simulation Parameters: $P = 4,\ K = 4,\ Q = 1$

Performance Analysis:

- **Baseline Comparison:** TxMF and ZFBF are interference-limited (saturation) at high SNR.
- **WSRBF-WMMSE Success:** Shows linear growth w.r.t. SNR, indicating effective **interference management**.
- **Initialization Impact: WMMSE2** (TxMF Init + 10 iter) captures most of the gain of **WMMSE1** (Random Init + Convergence), verifying the low-complexity advantage.
- **Role of Weights ($\mathbf{W}$):** The weight update acts as a **soft user selection**, nulling out users with poor channel compatibility to maximize sum-rate.

Code is available on GitHub

Thank You

Q & A

Derivation of Eq (3a)

$$\mathbf{x} = \sum_{k=1}^{K} \mathbf{B}_k \mathbf{d}_k, \quad \mathbf{x}^{\mathrm{H}}\mathbf{x} = \left(\sum_{j=1}^{K} \mathbf{d}_j^{\mathrm{H}} \mathbf{B}_j^{\mathrm{H}}\right) \left(\sum_{k=1}^{K} \mathbf{B}_k \mathbf{d}_k\right) = \sum_{j=1}^{K}\sum_{k=1}^{K} \mathbf{d}_j^{\mathrm{H}} \mathbf{B}_j^{\mathrm{H}} \mathbf{B}_k \mathbf{d}_k$$

As $\mathbf{d}_k^{\mathrm{H}} \mathbf{B}_k^{\mathrm{H}} \mathbf{B}_k \mathbf{d}_k$ is scalar, Tr can be applied:

$$\mathbf{d}_j^{\mathrm{H}} \mathbf{B}_j^{\mathrm{H}} \mathbf{B}_k \mathbf{d}_k = \mathrm{Tr}(\mathbf{B}_k^{\mathrm{H}} \mathbf{B}_k \mathbf{d}_k \mathbf{d}_k^{\mathrm{H}}), \quad \mathbb{E}[\mathbf{d}_j \mathbf{d}_k^{\mathrm{H}}] = \begin{cases} \mathbf{I} & \text{if } j = k \\ 0 & \text{if } j \neq k \end{cases}$$

Therefore,

$$\mathbb{E}[\mathbf{x}^{\mathrm{H}}\mathbf{x}] = \sum_{j=1}^{K}\sum_{k=1}^{K} \mathrm{Tr}(\mathbf{B}_j^{\mathrm{H}} \mathbf{B}_k \mathbb{E}[\mathbf{d}_k \mathbf{d}_j^{\mathrm{H}}]) = \sum_{k=1}^{K} \mathrm{Tr}(\mathbf{B}_k \mathbf{B}_k^{\mathrm{H}})$$

Return

Derivation of Eq (7)

$$J = \mathbb{E}\left[\left\|\mathbf{A}_k\mathbf{y}_k - \mathbf{d}_k\right\|^2\right] = \mathbb{E}\left[\mathrm{Tr}\left(\left(\mathbf{A}_k\mathbf{y}_k - \mathbf{d}_k\right)\left(\mathbf{A}_k\mathbf{y}_k - \mathbf{d}_k\right)^{\mathrm{H}}\right)\right]$$
$$= \mathbb{E}\left[\mathrm{Tr}\left(\left(\mathbf{A}_k\mathbf{y}_k - \mathbf{d}_k\right)\left(\mathbf{y}_k^{\mathrm{H}}\mathbf{A}_k^{\mathrm{H}} - \mathbf{d}_k^{\mathrm{H}}\right)\right)\right]$$

Let Correlation $\mathbf{R}_{yy} = \mathbb{E}\left[\mathbf{y}_k\mathbf{y}_k^{\mathrm{H}}\right]$, $\mathbf{R}_{dy} = \mathbb{E}\left[\mathbf{d}_k\mathbf{y}_k^{\mathrm{H}}\right]$, $\mathbf{R}_{yd} = \mathbb{E}\left[\mathbf{y}_k\mathbf{d}_k^{\mathrm{H}}\right] = \mathbf{R}_{dy}^{\mathrm{H}}$

$$J = \mathrm{Tr}\left(\mathbf{A}_k\mathbf{R}_{yy}\mathbf{A}_k^{\mathrm{H}} - \mathbf{A}_k\mathbf{R}_{yd} - \mathbf{R}_{dy}\mathbf{A}_k^{\mathrm{H}} + \mathbb{E}\left[\mathbf{d}_k\mathbf{d}_k^{\mathrm{H}}\right]\right)$$

Using matrix calculus,

- $\frac{\partial}{\partial \mathbf{X}^*}\mathrm{Tr}\left(\mathbf{X}\mathbf{R}\mathbf{X}^H\right) = \mathbf{X}\mathbf{R}$
- $\frac{\partial}{\partial \mathbf{X}^*}\cdot\mathrm{Tr}\left(\mathbf{A}\mathbf{X}^H\right) = \mathbf{A}$

$$\frac{\partial J}{\partial \mathbf{A}_k^*} = \mathbf{A}_k\mathbf{R}_{yy} - \mathbf{R}_{dy} = 0$$
$$\mathbf{A}_k^{\mathrm{MMSE}} = \mathbf{R}_{dy}\mathbf{R}_{yy}^{-1}$$

Derivation of Eq (7)

$$\begin{aligned}
\mathbf{R}_{dy} &= \mathbb{E}\left[\mathbf{d}_k \mathbf{y}_k^{\mathrm{H}}\right] \\
&= \mathbb{E}\left[\mathbf{d}_k \left(\mathbf{H}_k \mathbf{B}_k \mathbf{d}_k + \tilde{\mathbf{v}}_k\right)^{\mathrm{H}}\right] \\
&= \mathbb{E}\left[\mathbf{d}_k \mathbf{d}_k^{\mathrm{H}} \mathbf{B}_k^{\mathrm{H}} \mathbf{H}_k^{\mathrm{H}}\right] = \mathbb{E}\left[\mathbf{d}_k \mathbf{d}_k^{\mathrm{H}}\right] \mathbf{B}_k^{\mathrm{H}} \mathbf{H}_k^{\mathrm{H}} \\
&= \mathbf{B}_k^{\mathrm{H}} \mathbf{H}_k^{\mathrm{H}}
\end{aligned}$$

$$\begin{aligned}
\mathbf{R}_{yy} &= \mathbb{E}\left[\mathbf{y}_k \mathbf{y}_k^{\mathrm{H}}\right] \\
&= \mathbb{E}\left[\left(\mathbf{H}_k \mathbf{B}_k \mathbf{d}_k + \tilde{\mathbf{v}}_k\right)\left(\mathbf{H}_k \mathbf{B}_k \mathbf{d}_k + \tilde{\mathbf{v}}_k\right)^{\mathrm{H}}\right] \\
&= \mathbf{B}_k \mathbf{H}_k \mathbb{E}\left[\mathbf{d}_k \mathbf{d}_k^{\mathrm{H}}\right] \mathbf{B}_k^{\mathrm{H}} \mathbf{H}_k^{\mathrm{H}} + \mathbb{E}\left[\tilde{\mathbf{v}}_k \tilde{\mathbf{v}}_k^{\mathrm{H}}\right] \\
&= \mathbf{B}_k \mathbf{H}_k \mathbf{B}_k^{\mathrm{H}} \mathbf{H}_k^{\mathrm{H}} + \mathbf{R}_{\tilde{v}_k \tilde{v}_k}
\end{aligned}$$

Therefore,

$$\mathbf{A}_k^{\mathrm{MMSE}} = \underbrace{\mathbf{B}_k^{\mathrm{H}} \mathbf{H}_k^{\mathrm{H}}}_{\mathbf{R}_{dy}} \underbrace{\left(\mathbf{H}_k \mathbf{B}_k \mathbf{B}_k^{\mathrm{H}} \mathbf{H}_k^{\mathrm{H}} + \mathbf{R}_{\tilde{v}_k \tilde{v}_k}\right)^{-1}}_{\mathbf{R}_{yy}^{-1}}$$

Derivation of Eq (8)

$$
\begin{aligned}
\mathbf{E}_k &= \mathbb{E}\left[\left(\mathbf{A}_k^{\mathrm{MMSE}}\mathbf{y}_k - \mathbf{d}_k\right)\left(\mathbf{A}_k^{\mathrm{MMSE}}\mathbf{y}_k - \mathbf{d}_k\right)^{\mathrm{H}}\right] \\
&= \mathbf{A}_k\mathbf{R}_{yy}\mathbf{A}_k^{\mathrm{H}} - \mathbf{A}_k\mathbf{R}_{yd} - \mathbf{R}_{dy}\mathbf{A}_k^{\mathrm{H}} + \mathbf{R}_{dd} \\
&= (\mathbf{R}_{dy}\mathbf{R}_{yy}^{-1})\mathbf{R}_{yy}\mathbf{A}_k^{\mathrm{H}} - (\mathbf{R}_{dy}\mathbf{R}_{yy}^{-1})\mathbf{R}_{yd} - \mathbf{R}_{dy}\mathbf{A}_k^{\mathrm{H}} + \mathbf{R}_{dd} \\
&= \mathbf{R}_{dd} - \mathbf{R}_{dy}\mathbf{R}_{yy}^{-1}\mathbf{R}_{yd} \\
&= \mathbf{I} - \mathbf{B}_k^{\mathrm{H}}\mathbf{H}_k^{\mathrm{H}}\left(\mathbf{B}_k\mathbf{H}_k\mathbf{B}_k^{\mathrm{H}}\mathbf{H}_k^{\mathrm{H}} + \mathbf{R}_{\tilde{v}_k\tilde{v}_k}\right)^{-1}\mathbf{H}_k\mathbf{B}_k
\end{aligned}
$$

Apply Woodbury matrix identity $\mathbf{I} - \mathbf{U}(\mathbf{C} + \mathbf{V}\mathbf{U})^{-1}\mathbf{V} = \left(\mathbf{I} + \mathbf{U}\mathbf{C}^{-1}\mathbf{V}\right)^{-1}$,
Let $\mathbf{U} = \mathbf{B}_k^{\mathrm{H}}\mathbf{H}_k^{\mathrm{H}}$, $\quad \mathbf{V} = \mathbf{H}_k\mathbf{B}_k$, $\quad \mathbf{C} = \mathbf{R}_{\tilde{v}_k\tilde{v}_k}$

$$
\begin{aligned}
\mathbf{E}_k &= \left(\mathbf{I} + \left(\mathbf{B}_k^{\mathrm{H}}\mathbf{H}_k^{\mathrm{H}}\right)\left(\mathbf{R}_{\bar{v}_k\bar{v}_k}\right)^{-1}\left(\mathbf{H}_k\mathbf{B}_k\right)\right)^{-1} \\
&= \left(\mathbf{I} + \mathbf{B}_k^{\mathrm{H}}\mathbf{H}_k^{\mathrm{H}}\mathbf{R}_{\bar{v}_k\bar{v}_k}^{-1}\mathbf{H}_k\mathbf{B}_k\right)^{-1}
\end{aligned}
$$

Return

Derivation of Eq (14)

$$\frac{\partial \log\det \mathbf{K}}{\partial x} = \mathrm{Tr}(\mathbf{K}^{-1}\frac{\partial \mathbf{K}}{\partial x})$$

$$\frac{\partial R_k}{\partial [\mathbf{B}_k^*]_{nm}} = \mathrm{Tr}\left(\mathbf{K}^{-1}\frac{\partial \mathbf{K}}{\partial [\mathbf{B}_k^*]_{nm}}\right) = \mathrm{Tr}\left(\mathbf{E}_k\frac{\partial \mathbf{K}}{\partial [\mathbf{B}_k^*]_{nm}}\right)$$

$$\frac{\partial \mathbf{K}}{\partial [\mathbf{B}_k^*]_{nm}} = \mathbf{e}_m\mathbf{e}_n^{\mathrm{T}}\mathbf{C}\mathbf{B}_k$$

$$\text{Target: } R_k = \log\det(\mathbf{K})$$

$$\text{Substitution: } \mathbf{K} = \mathbf{I} + \mathbf{B}_k^{\mathrm{H}}\mathbf{C}\mathbf{B}_k$$

$$\text{Constant Matrix: } \mathbf{C} = \mathbf{H}_k^{\mathrm{H}}\mathbf{R}_{\tilde{v}_k}^{-1}\mathbf{H}_k$$

$$\text{MMSE Matrix: } \mathbf{E}_k = \mathbf{K}^{-1}$$

$$\begin{aligned}\frac{\partial R_k}{\partial [\mathbf{B}_k^*]_{nm}} &= \mathrm{Tr}\left(\mathbf{E}_k\mathbf{e}_m\mathbf{e}_n^{\mathrm{T}}\mathbf{C}\mathbf{B}_k\right)\\ &= \mathrm{Tr}\left(\mathbf{e}_n^{\mathrm{T}}\mathbf{C}\mathbf{B}_k\mathbf{E}_k\mathbf{e}_m\right)\\ &= [\mathbf{C}\mathbf{B}_k\mathbf{E}_k]_{nm}\end{aligned}$$

$$\nabla_{\mathbf{B}_k} R_k = \mathbf{C}\mathbf{B}_k\mathbf{E}_k = \mathbf{H}_k^{\mathrm{H}}\mathbf{R}_{\tilde{v}_k\tilde{v}_k}^{-1}\mathbf{H}_k\mathbf{B}_k\mathbf{E}_k$$

Return

Derivation of Eq (17)

$$\mathbf{S} = \mathbf{X} + \mathbf{L}\mathbf{K}\mathbf{C}\mathbf{C}^{\mathrm{H}}\mathbf{K}^{\mathrm{H}}\mathbf{L}^{\mathrm{H}}$$

$$\nabla_{\mathbf{K}} h = \mathbf{L}^{\mathrm{H}} \left(\nabla_{\mathbf{S}} h\right) \mathbf{L}\mathbf{K}\mathbf{C}\mathbf{C}^{\mathrm{H}}$$

$$R_i = \log\det\left(\mathbf{I} + \mathbf{B}_i^{\mathrm{H}}\mathbf{H}_i^{\mathrm{H}}\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1}\mathbf{H}_i\mathbf{B}_i\right)$$

$$\mathbf{R}_{\tilde{v}_i\tilde{v}_i} = \mathbf{I}_i + \sum_{k=1, k\neq i}^{K} \mathbf{H}_i\mathbf{B}_k\mathbf{B}_k^{\mathrm{H}}\mathbf{H}_i^{\mathrm{H}}$$

$$\mathbf{R}_{\tilde{v}_i\tilde{v}_i} = \underbrace{\left(\mathbf{I}_i + \sum_{\substack{j=1 \\ j\neq i,k}}^{K} \mathbf{H}_i\mathbf{B}_j\mathbf{B}_j^{\mathrm{H}}\mathbf{H}_i^{\mathrm{H}}\right)}_{\mathbf{X}\ \text{(constant)}} + \mathbf{H}_i\mathbf{B}_k\mathbf{I}\mathbf{B}_k^{\mathrm{H}}\mathbf{H}_i^{\mathrm{H}}$$

$$h = R_i\ ,\ \mathbf{S} = \mathbf{R}_{\tilde{v}_i\tilde{v}_i}\ ,\ \mathbf{L} = \mathbf{H}_i\ ,\ \mathbf{K} = \mathbf{B}_k\ ,\ \mathbf{C} = \mathbf{C}^{\mathrm{H}} = \mathbf{I}$$

$$\nabla_{\mathbf{B}_k} R_i = \mathbf{H}_i^{\mathrm{H}} \left(\nabla_{\mathbf{R}_{\tilde{v}_i\tilde{v}_i}} R_i\right) \mathbf{H}_i\mathbf{B}_k \tag{15}$$

Derivation of Eq (17)

1. Definitions

$$R_i = \log\det(\mathbf{K}_i)$$

$$\mathbf{K}_i = \mathbf{I} + \mathbf{A}_i^{\mathrm{H}}\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1}\mathbf{A}_i \quad (\text{where} \mathbf{A}_i = \mathbf{H}_i\mathbf{B}_i)$$

$$\mathbf{E}_i = \mathbf{K}_i^{-1} \quad (\text{MMSE Matrix})$$

2. Differential of Log-Det (Chain Rule)

$$dR_i = \mathrm{Tr}(\mathbf{K}_i^{-1}\, d\mathbf{K}_i) = \mathrm{Tr}(\mathbf{E}_i d\mathbf{K}_i)$$

3. Differential of Inverse Matrix

Using $d(\mathbf{X}^{-1}) = -\mathbf{X}^{-1}(d\mathbf{X})\mathbf{X}^{-1}$:

$$\begin{aligned} d\mathbf{K}_i &= \mathbf{A}_i^{\mathrm{H}}\, d(\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1})\mathbf{A}_i \\ &= \mathbf{A}_i^{\mathrm{H}}\left(-\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1}(d\mathbf{R}_{\tilde{v}_i\tilde{v}_i})\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1}\right)\mathbf{A}_i \end{aligned}$$

4. Substitution & Trace Cyclic Property

$$dR_i = \mathrm{Tr}\left(\mathbf{E}_i\left[-\mathbf{A}_i^{\mathrm{H}}\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1}(d\mathbf{R}_{\tilde{v}_i\tilde{v}_i})\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1}\mathbf{A}_i\right]\right)$$

Move $(d\mathbf{R}_{\tilde{v}_i\tilde{v}_i})$ to the end:

$$dR_i = \mathrm{Tr}\left(\underbrace{\left[-\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1}\mathbf{A}_i\mathbf{E}_i\mathbf{A}_i^{\mathrm{H}}\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1}\right]}_{\text{Gradient}}\, d\mathbf{R}_{\tilde{v}_i\tilde{v}_i}\right)$$

5. Obtain Eq (16)

$$\nabla_{\mathbf{R}_{\tilde{v}_i\tilde{v}_i}} R_i = -\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1}\mathbf{H}_i\mathbf{B}_i\mathbf{E}_i\mathbf{B}_i^{\mathrm{H}}\mathbf{H}_i^{\mathrm{H}}\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1}$$

Derivation of Eq (17)

Combining Eq (15) and (16), we obtain Eq (17):

$$
\begin{aligned}
\nabla_{\mathbf{R}_{\tilde{v}_i\tilde{v}_i}} R_i &= -\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1}\mathbf{H}_i\mathbf{B}_i\mathbf{E}_i\mathbf{B}_i^{\mathrm{H}}\mathbf{H}_i^{\mathrm{H}}\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1} && (16)\\
\nabla_{\mathbf{B}_k} R_i &= \mathbf{H}_i^{\mathrm{H}}\left(\nabla_{\mathbf{R}_{\tilde{v}_i\tilde{v}_i}} R_i\right)\mathbf{H}_i\mathbf{B}_k && (15)\\
&= -\mathbf{H}_i^{\mathrm{H}}\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1}\mathbf{H}_i\mathbf{B}_i\mathbf{E}_i\mathbf{B}_i^{\mathrm{H}}\mathbf{H}_i^{\mathrm{H}}\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1}\mathbf{H}_i\mathbf{B}_k && (17)
\end{aligned}
$$

Return

Derivation of Eq (20)

$$\mathbf{E}_k = \left(\mathbf{I}_k + \mathbf{B}_k^{\mathrm{H}} \underbrace{\mathbf{H}_k^{\mathrm{H}} \mathbf{R}_{\tilde{v}_k \tilde{v}_k}^{-1} \mathbf{H}_k}_{\mathbf{F}} \mathbf{B}_k \right)^{-1}$$

$$\begin{aligned} d\left(\mathbf{X}^{-1}\right) &= -\mathbf{X}^{-1} \left(d\mathbf{X}\right) \mathbf{X}^{-1}, \\ d\left(\mathbf{E}_k\right) &= -\mathbf{E}_k \left(d\mathbf{E}_k^{-1}\right) \mathbf{E}_k \\ &= -\mathbf{E}_k \left(d\left(\mathbf{I}_k + \mathbf{B}_k^{\mathrm{H}} \mathbf{F} \mathbf{B}_k\right)\right) \mathbf{E}_k \ , \\ d\left(\mathbf{B}_k^{\mathrm{H}} \mathbf{F} \mathbf{B}_k\right) &= \left(d\mathbf{B}_k^{\mathrm{H}}\right) \mathbf{F} \mathbf{B}_k \end{aligned}$$

$$\begin{aligned} d\,\mathrm{Tr}\left(\mathbf{W}_k \mathbf{E}_k\right) &= \mathrm{Tr}\left(\mathbf{W}_k \left(d\mathbf{E}_k\right)\right) \\ &= \mathrm{Tr}\left(\mathbf{W}_k \left(-\mathbf{E}_k \left(d\mathbf{B}_k^{\mathrm{H}}\right) \mathbf{F} \mathbf{B}_k \mathbf{E}_k\right)\right) \\ &= -\mathrm{Tr}\left(\mathbf{W}_k \mathbf{E}_k \left(d\mathbf{B}_k^{\mathrm{H}}\right) \mathbf{F} \mathbf{B}_k \mathbf{E}_k\right) \\ &= -\mathrm{Tr}\left(\underbrace{\mathbf{F} \mathbf{B}_k \mathbf{E}_k \mathbf{W}_k \mathbf{E}_k}_{\nabla_{\mathbf{B}_k} \mathrm{Tr}\left(\mathbf{W}_k \mathbf{E}_k\right)} \left(d\mathbf{B}_k^{\mathrm{H}}\right) \right) \end{aligned}$$

$$\therefore \nabla_{\mathbf{B}_k} \mathrm{Tr}\left(\mathbf{W}_k \mathbf{E}_k\right) = -\mathbf{H}_k^{\mathrm{H}} \mathbf{R}_{\tilde{v}_k \tilde{v}_k}^{-1} \mathbf{H}_k \mathbf{B}_k \mathbf{E}_k \mathbf{W}_k \mathbf{E}_k \tag{20}$$

Return

Derivation of Eq (21)

$$\mathbf{E}_i = \left(\mathbf{I}_i + \mathbf{B}_i^{\mathrm{H}}\mathbf{H}_i^{\mathrm{H}}\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1}\mathbf{H}_i\mathbf{B}_i\right)^{-1}$$

$\mathbf{B}_k$is only relevant with $\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1}$

$$\begin{aligned} d\left(\mathbf{E}_i\right) &= -\mathbf{E}_i\left(d\mathbf{E}_i^{-1}\right)\mathbf{E}_i \\ &= -\mathbf{E}_i\left(d\left(\mathbf{I}_k + \mathbf{B}_i^{\mathrm{H}}\mathbf{H}_i^{\mathrm{H}}\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1}\mathbf{H}_i\mathbf{B}_i\right)\right)\mathbf{E}_i \\ &= -\mathbf{E}_i\left(\mathbf{B}_i^{\mathrm{H}}\mathbf{H}_i^{\mathrm{H}}\left(d\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1}\right)\mathbf{H}_i\mathbf{B}_i\right)\mathbf{E}_i \end{aligned}$$

$$d\left(\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1}\right) = -\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1}\left(d\mathbf{R}_{\tilde{v}_i\tilde{v}_i}\right)\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1}$$

$$\begin{aligned} d\,\mathrm{Tr}\left(\mathbf{W}_i\mathbf{E}_i\right) &= \mathrm{Tr}\left(\mathbf{W}_i\left(d\mathbf{E}_i\right)\right) \\ &= -\mathrm{Tr}\left(\mathbf{W}_i\mathbf{E}_i\mathbf{B}_i^{\mathrm{H}}\mathbf{H}_i^{\mathrm{H}}\left(d\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1}\right)\mathbf{H}_i\mathbf{B}_i\mathbf{E}_i\right) \\ &= -\mathrm{Tr}\left(\mathbf{H}_i\mathbf{B}_i\mathbf{E}_i\mathbf{W}_i\mathbf{E}_i\mathbf{B}_i^{\mathrm{H}}\mathbf{H}_i^{\mathrm{H}}\left(d\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1}\right)\right) \\ &= \mathrm{Tr}\left(\mathbf{H}_i\mathbf{B}_i\mathbf{E}_i\mathbf{W}_i\mathbf{E}_i\mathbf{B}_i^{\mathrm{H}}\mathbf{H}_i^{\mathrm{H}}\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1}\left(d\mathbf{R}_{\tilde{v}_i\tilde{v}_i}\right)\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1}\right) \\ &= \mathrm{Tr}\left(\underbrace{\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1}\mathbf{H}_i\mathbf{B}_i\mathbf{E}_i\mathbf{W}_i\mathbf{E}_i\mathbf{B}_i^{\mathrm{H}}\mathbf{H}_i^{\mathrm{H}}\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1}}_{\nabla_{\mathbf{R}_{\tilde{v}_i\tilde{v}_i}}\mathrm{Tr}(\mathbf{W}_i\mathbf{E}_i)}\left(d\mathbf{R}_{\tilde{v}_i\tilde{v}_i}\right)\right) \end{aligned}$$

Derivation of Eq (21)

$$\mathbf{R}_{\tilde{v}_i\tilde{v}_i} = \cdots + \mathbf{H}_i\mathbf{B}_k\mathbf{B}_k^{\mathrm{H}}\mathbf{H}_i^{\mathrm{H}} + \cdots$$

$$d\mathbf{R}_{\tilde{v}_i\tilde{v}_i} = \mathbf{H}_i\mathbf{B}_k\left(d\mathbf{B}_k^{\mathrm{H}}\right)\mathbf{H}_i^{\mathrm{H}}$$

$$\begin{aligned} d\,\mathrm{Tr}\left(\mathbf{W}_i\mathbf{E}_i\right) &= \mathrm{Tr}\left(\alpha\left(d\mathbf{R}_{\tilde{v}_i\tilde{v}_i}\right)\right) \\ &= \mathrm{Tr}\left(\alpha\mathbf{H}_i\mathbf{B}_k\left(d\mathbf{B}_k^{\mathrm{H}}\right)\mathbf{H}_i^{\mathrm{H}}\right) \\ &= \mathrm{Tr}\left(\underbrace{\mathbf{H}_i^{\mathrm{H}}\alpha\mathbf{H}_i\mathbf{B}_k}_{\nabla_{\mathbf{B}_k}\mathrm{Tr}(\mathbf{W}_i\mathbf{E}_i)}\left(d\mathbf{B}_k^{\mathrm{H}}\right)\right) \end{aligned}$$

$$\nabla_{\mathbf{B}_k}\mathrm{Tr}\left(\mathbf{W}_i\mathbf{E}_i\right) = \mathbf{H}_i^{\mathrm{H}}\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1}\mathbf{H}_i\mathbf{B}_i\mathbf{E}_i\mathbf{W}_i\mathbf{E}_i\mathbf{B}_i^{\mathrm{H}}\mathbf{H}_i^{\mathrm{H}}\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1}\mathbf{H}_i\mathbf{B}_k \tag{21}$$

Return

Derivation of Eq (23)

Problem Formulation: Minimize the Weighted MSE subject to the transmit power constraint E_{tx}. Let $\mathbf{B} = b\bar{\mathbf{B}}$, where b is a scaling factor.

$$\begin{aligned} \min_{\bar{\mathbf{B}}, b} \quad & J = \mathbb{E}\left[\left\|\mathbf{W}^{1/2}(\mathbf{s} - b^{-1}\mathbf{A}\mathbf{y})\right\|^2\right] \\ \text{s.t.} \quad & \mathrm{Tr}(\mathbf{B}\mathbf{B}^{\mathrm{H}}) = b^2 \mathrm{Tr}(\bar{\mathbf{B}}\bar{\mathbf{B}}^{\mathrm{H}}) = E_{tx} \end{aligned}$$

Step 1: Constraint Substitution

From the constraint, we substitute the scaling factor b:

$$b^{-2} = \frac{\mathrm{Tr}(\bar{\mathbf{B}}\bar{\mathbf{B}}^{\mathrm{H}})}{E_{tx}}$$

Step 2: Cost Function Expansion

Substituting $\mathbf{y} = \mathbf{H}b\bar{\mathbf{B}}\mathbf{s} + \mathbf{n}$ and b^{-2} into the cost function J:

$$\begin{aligned} J(\bar{\mathbf{B}}) &= \mathrm{Tr}\left(\mathbf{W}(\mathbf{I} - \mathbf{A}\mathbf{H}\bar{\mathbf{B}})(\mathbf{I} - \mathbf{A}\mathbf{H}\bar{\mathbf{B}})^{\mathrm{H}}\right) + b^{-2}\mathrm{Tr}(\mathbf{W}\mathbf{A}\mathbf{A}^{\mathrm{H}}) \\ &= \underbrace{\mathrm{Tr}\left(\mathbf{W}(\mathbf{I} - \mathbf{A}\mathbf{H}\bar{\mathbf{B}})(\dots)^{\mathrm{H}}\right)}_{\text{Error Term}} + \underbrace{\frac{\mathrm{Tr}(\mathbf{W}\mathbf{A}\mathbf{A}^{\mathrm{H}})}{E_{tx}}\mathrm{Tr}(\bar{\mathbf{B}}\bar{\mathbf{B}}^{\mathrm{H}})}_{\text{Penalty Term (from Noise)}} \end{aligned}$$

Derivation of Eq (23)

Step 3: Complex Gradient

Taking the gradient with respect to $\bar{\mathbf{B}}^*$ and setting it to zero:

$$\frac{\partial J}{\partial \bar{\mathbf{B}}^*} = -\mathbf{H}^{\mathrm{H}}\mathbf{A}^{\mathrm{H}}\mathbf{W} + \mathbf{H}^{\mathrm{H}}\mathbf{A}^{\mathrm{H}}\mathbf{W}\mathbf{A}\mathbf{H}\bar{\mathbf{B}} + \underbrace{\frac{\mathrm{Tr}(\mathbf{W}\mathbf{A}\mathbf{A}^{\mathrm{H}})}{E_{tx}}\mathbf{I}_P}_{\text{Regularization}}\bar{\mathbf{B}} = 0$$

Step 4: Solution

Rearranging the terms to solve for $\bar{\mathbf{B}}$:

$$\left(\mathbf{H}^{\mathrm{H}}\mathbf{A}^{\mathrm{H}}\mathbf{W}\mathbf{A}\mathbf{H} + \frac{\mathrm{Tr}(\mathbf{W}\mathbf{A}\mathbf{A}^{\mathrm{H}})}{E_{tx}}\mathbf{I}_P\right)\bar{\mathbf{B}} = \mathbf{H}^{\mathrm{H}}\mathbf{A}^{\mathrm{H}}\mathbf{W}$$

Finally, we obtain the structure in (23):

$$\bar{\mathbf{B}} = \left(\mathbf{H}^{\mathrm{H}}\mathbf{A}^{\mathrm{H}}\mathbf{W}\mathbf{A}\mathbf{H} + \frac{\mathrm{Tr}(\mathbf{W}\mathbf{A}\mathbf{A}^{\mathrm{H}})}{E_{tx}}\mathbf{I}_P\right)^{-1}\mathbf{H}^{\mathrm{H}}\mathbf{A}^{\mathrm{H}}\mathbf{W}$$

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